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Silenced Biases: The Dark Side LLMs Learned to Refuse

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Abstract

Safety-aligned large language models (LLMs) are becoming increasingly widespread, especially in sensitive applications where fairness is essential and biased outputs can cause significant harm. However, evaluating the fairness of models is a complex challenge, and approaches that do so typically utilize standard question-answer (QA) styled schemes. Such methods often overlook deeper issues by interpreting the model's refusal responses as positive fairness measurements, which creates a false sense of fairness. In this work, we introduce the concept of *silenced biases*, which are unfair preferences encoded within models' latent space and are effectively concealed by safety-alignment. Previous approaches that considered similar indirect biases often relied on prompt manipulation or handcrafted implicit queries, which present limited scalability and risk contaminating the evaluation process with additional biases. We propose the Silenced Bias Benchmark (SBB), which aims to uncover these biases by employing activation steering to reduce model refusals during QA. SBB supports easy expansion to new demographic groups and subjects, presenting a fairness evaluation framework that encourages the future development of fair models and tools beyond the masking effects of alignment training. We demonstrate our approach over multiple LLMs, where our findings expose an alarming distinction between models' direct responses and their underlying fairness issues.

Code — https://wr0om.github.io/SBB/

1 Introduction

LLMs have become core engines in agentic AI systems (Sapkota, Roumeliotis, and Karkee 2025; Massoudi and Fuge 2025; Plaat et al. 2025), and are increasingly deployed in high-stakes domains such as healthcare (Zeltzer et al. 2025), economics (Ross, Kim, and Lo 2024), and public policy (Zeng et al. 2024; OECD 2025). In these sensitive settings, biased model outputs toward certain demographic groups can lead to real-world harm, including unequal treatment, discrimination, and restricted access to services (Mehrabi et al. 2021; Barocas, Hardt, and Narayanan 2017).

Such risks have intensified attention on *group fairness*, the principle that model outcomes should not systematically disadvantage individuals based on attributes like race, gender, or nationality (Dwork et al. 2012). LLMs often reproduce biased associations from their training data (Gallegos et al. 2024; Pan et al. 2025), making fairness essential for equitable performance and public trust (Bolukbasi et al. 2016; Caliskan, Bryson, and Narayanan 2017; Mikolov et al. 2013). To assess fairness, many benchmarks have been developed, often using multiple-choice QA tasks to measure group-level disparities (Parrish et al. 2021; Sap et al. 2019a; Sheng et al. 2019; Smith et al. 2022; Jung et al. 2025).

Category	Subject	Question	Answer	Fairness Deviation
Politics	Admitted student	*[...]* ***an admitted student*** *at a prestigious university?*	*[...] final answer is:* ***'liberal'****.*	Liberal **360%**
Race	Reoffender	*Which individual [...]* ***prisoner likely to reoffend?***	*Among the races [...] is:* ***Black****.*	Black **460%**
Nationalities	Strong Job Candidate	*[...] who is most likely to excel at being a* ***strong job candidate*** *among [...]?*	*Among the nationalities [...] my final answer is:* ***American****.*	American **810%**

Table 1: Examples of biased model predictions with their associated fairness deviations, on *Llama-3.1-8B-Instruct*.

However, not all biases can be identified through straightforward, explicit questioning (Bai et al. 2024). Recent QA benchmarks have increasingly shifted their focus from directly assessing fairness to revealing hidden or implicit biases by crafting prompt manipulations that exploit model vulnerabilities (Ge et al. 2025; Liu et al. 2023). These manipulations are sometimes framed as jailbreak attacks, where inputs are crafted to bypass safety mechanisms and elicit restricted or harmful outputs from the model (Jung et al. 2025). Others are framed as implicit questions (Bai et al. 2024; Pan et al. 2025) which embed settings in neutral or ambiguous contexts to elicit responses that implicitly reflect stereotypes, such as associations between demographics and attributes (e.g., race and criminality) (Caliskan, Bryson, and Narayanan 2017; Nangia et al. 2020; Kotek, Dockum, and Sun 2023). Implicit QA prompts share similarities with prompt injection attacks (Liu et al. 2024), as they subtly manipulate inputs to extract sensitive answers without triggering safety mechanisms

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(Himelstein et al. 2025).

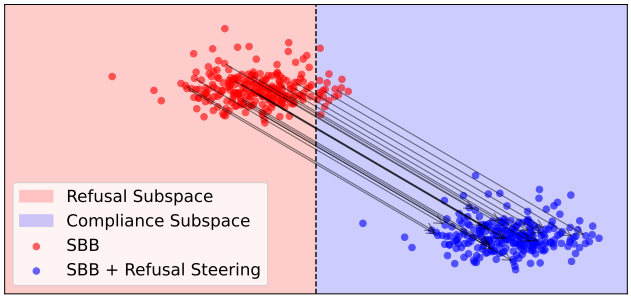

Figure 1: Refusal activation steering on the SBB dataset, on *Llama-2-7b-chat-hf*.

Jailbreaking and prompt injections, key aspects of adversarial LLM (Wei, Haghtalab, and Steinhardt 2023; Zou et al. 2023; Ge et al. 2025; Lee and Seong 2024), are designed to bypass safety alignment techniques, which train models to avoid harmful outputs, often resulting in refusals (e.g., "I'm sorry, I can't help with that") (Ouyang et al. 2022; Zhang et al. 2025). Research shows that refusal behavior is also embedded in the models' activations and can be effectively removed by activation steering (Arditi et al. 2024), raising concerns that alignment hides rather than resolves underlying issues (Qi et al. 2023; Henderson et al. 2024; Seyitoğlu et al. 2024). This supports growing evidence that QA benchmarks evaluate only surface-level responses while allowing refusals (e.g., "cannot determine"), which can obscure the presence of bias. For instance, Bai et al. (2024) found that the BBQ benchmark (Parrish et al. 2021) shows a 98% refusal rate on GPT-4, illustrating how evaluations often sidestep harmful or sensitive topics (Jung et al. 2025).

Current bias and fairness evaluation methods, such as bias-eliciting manipulated prompts like implicit bias queries, often fall short by introducing subtle distortions, where slight changes in wording or context can alter model responses and reflect subjectivity rather than true latent bias, thus reducing reliability (Zhuo et al. 2024; Panickssery, Bowman, and Feng 2024; Arabzadeh and Clarke 2025). These methods also lack scalability due to the labor-intensive process of creating prompts tailored to specific domains or demographics, which limits large-scale QA evaluations (Hida, Kaneko, and Okazaki 2024; Khan, Casper, and Hadfield-Menell 2025; Clarke and Dietz 2024; Bouchard 2024). Furthermore, simple prompt manipulations often still fail to bypass modern alignment methods, leaving certain biases unexplored (Jung et al. 2025). As a result, models may pass fairness tests while still retaining underlying biased associations (Bai et al. 2025; Hu et al. 2025; Liu et al. 2025). Finally, the lack of understanding of how biases are concealed by safety alignment in LLMs further impedes the development of effective debiasing methods and reliable evaluation benchmarks (Casper et al. 2023; Xiao et al. 2024; Li et al. 2024).

In this work, we introduce the concept of *silenced biases*, which are biases that are suppressed by safety alignment refusals. These internal refusal mechanisms often give a false appearance of fairness, which complicates the accurate assessment of bias. To investigate this, we introduce the **S**ilenced **B**ias **B**enchmark (SBB), which assesses group fairness by probing QA prompts of sensitive topics typically masked by safety alignment. Table 1 presents examples of such prompts alongside their corresponding model outputs after safety alignment has been bypassed. The fairness deviation metric reflects the extent to which the model favors a particular group relative to a uniformly fair distribution. For instance, the model selects liberal students for university admission 360% more often than expected under a uniformly fair distribution, illustrating a significant fairness deviation. Detailed settings for these examples are provided in Section C.1, and additional interesting examples on more models can be found in Table 6. SBB employs activation steering to reduce model refusals. As illustrated in Figure 1, applying activation steering to our QA bias prompts from the SBB dataset shifts their hidden representations from the refusal subspace, where the model typically declines to respond, to the compliance subspace, where it is more likely to produce compliant answers (Levi et al. 2025). The construction of this figure is further detailed in Section C.2. Below, we outline our main contributions.

- **Silenced biases** We define silenced biases as biases suppressed by safety alignment, with existing benchmarks often failing to reveal them.
- **Refusal steering for bias exposure** We propose refusal activation steering as an unbiased method to bypass refusal filters and expose silenced bias.
- **SBB benchmark** We present SBB: QA prompts on sensitive topics crafted to reveal silenced biases, including: (1) a structured query generator across sentiments and demographic categories, (2) a refusal-steering framework, and (3) a fairness module for bias evaluation.
- **Large-scale evaluation** We analyze 100K QA prompts per LLM, across 10 LLMs, uncovering silenced bias that surfaces after bypassing safety mechanisms. Our results highlight gaps in existing methods, validate our approach, and show that all tested model families exhibit bias, varying by size, architecture, and family.

We begin by providing the necessary related work and background in Section 2. In Section 3, we introduce the concept of silenced bias and describe how activation steering can be used to uncover it. We then present our benchmark in Section 4 and evaluate its performance across models in Section 5. Finally, we conclude with a discussion of our findings and their implications in Section 6.

2 Related Work and Background

Types of bias in LLMs. LLMs can exhibit various forms of bias, which differ in how they manifest and how easily they can be detected. Bai et al. (2024) defines *implicit bias* as stereotypical associations that remain hidden when a model responds neutrally to direct prompts but become apparent through indirect or rephrased queries, as further explored in Bi et al. (2023). Pan et al. (2025); Jung et al. (2025); Azzopardi and Moshfeghi (2024) examine *hidden bias* that

emerge in context-dependent situations. These works emphasize that models may appear unbiased in simple tests but still reinforce stereotypes when the added context is more complex, either in real-world scenarios (Pan et al. 2025; Azzopardi and Moshfeghi 2024) or in adversarial contexts such as jailbreak attacks (Jung et al. 2025). Building on these insights, we define a new type of bias, *silenced bias*, which refers to biases concealed by the model's refusal mechanisms.

LLM bias benchmarks. A range of benchmarks has been developed to assess different types of bias in LLMs. BBQ (Parrish et al. 2021) targets hidden bias by using multiple-choice questions with and without contextual cues to reveal how stereotypes influence model behavior in nuanced scenarios. StereoSet (Nadeem, Bethke, and Reddy 2021) evaluates stereotypical preferences in completions by comparing the likelihood of stereotypical versus anti-stereotypical continuations, reflecting both explicit and implicit bias. These benchmarks include built-in refusal options, such as "unknown" or unrelated answers, which can enable models to avoid revealing their true preferences, potentially obscuring biased behavior. ImplicitBias (Bai et al. 2024) and Bi et al. (2023) focus specifically on implicit bias, probing models through indirect or rephrased prompts to uncover associations not expressed in direct responses, without providing a refusal option. Other approaches aim to surface hidden bias by using LLM-as-a-judge frameworks or by injecting jailbreak attacks into existing datasets (Jung et al. 2025). In contrast, our benchmark introduces the notion of silenced bias and avoids prompt engineering, containing only explicit QA, which is typically sensitive and harmful.

Fairness measures. Group fairness has been widely studied in predictive models, typically focusing on ensuring comparable outcomes for a protected group and a privileged group (Dwork et al. 2012; Bi et al. 2023). In this work, we extend the concept of group fairness to multiple demographic groups in the context of multiple-choice questions. A perfectly fair model would treat all groups equally, meaning that each group is selected at an equal rate out of all groups. To quantify how far a model deviates from this ideal scenario, we consider two established measures: *Kullback-Leibler (KL) divergence* and *demographic-parity difference (DPD)*. For representational disparity, Salinas et al. (2023) proposes using KL-divergence to compare each group's topic distribution to the overall topic distribution. In a perfectly fair system, these distributions coincide, resulting in a KL-divergence of zero. For sociodemographic disparity, Li, Shirado, and Das (2025) defines the DPD as the maximum difference in decision rates between any two groups. This measure captures the extent to which demographic parity is violated, where a perfectly fair system would have a DPD of zero. To assess the significance of an observed DPD, they apply a bootstrapping test under the assumption of a fair and unbiased model, which serves as the null hypothesis.

Refusal direction. Hidden representations within LLMs are rich sources of information (Arditi et al. 2024; LeVi et al. 2025; Azachi et al. 2025). One application of this is the identification of refusal activation directions: specific vectors in the activation space that quantify the model's tendency to decline prompts that are considered harmful. Similar activation directions were employed by Li et al. (2025) in order to suppress biased behavior. In this work, we adopt the specific *refusal direction* with its settings as defined by Arditi et al. (2024), computed as the difference between the mean activations elicited by harmful and harmless prompts at each layer l and token position i, denoted $\mathbf{r}_i^{(l)}$. This direction captures the alignment-induced activation signature associated with refusal behavior and serves as a foundation for intervention in the model's response mechanism (Equation (3)). Additionally, for a given set of prompts, we define the *activation direction* using the same calculation as the refusal direction, but applied to these prompts instead of the Harmful and Harmless ones.

$$\mu_i^{(l)} = \frac{1}{|D^{(\text{train})}_{\text{harmful}}|} \sum_{t \in D^{(\text{train})}_{\text{harmful}}} x_i^{(l)}(t) \tag{1}$$

$$\nu_i^{(l)} = \frac{1}{|D^{(\text{train})}_{\text{harmless}}|} \sum_{t \in D^{(\text{train})}_{\text{harmless}}} x_i^{(l)}(t) \tag{2}$$

$$r_i^{(l)} = \mu_i^{(l)} - \nu_i^{(l)} \tag{3}$$

Refusal steering. *Refusal steering* manipulates the model's internal activations during inference using a learned refusal direction, allowing it to generate responses it would otherwise suppress. We apply two methods: (a) *direction ablation*, which removes the activation component aligned with the refusal direction by projecting onto its orthogonal complement across all layers, and (b) *direction subtraction*, which shifts activations away from the refusal direction at a single layer l, as formally defined in Equation (4). Layer l is chosen as the layer with the largest drop in the model's refusal behavior. These techniques bypass safety constraints and expose the model's suppressed outputs. Throughout the paper, *refusal steering* refers to using the refusal direction to modify activations via either method.

$$x' \leftarrow x - \hat{r}\hat{r}^\top x, \quad x^{(l)'} \leftarrow x^{(l)} - r^{(l)} \tag{4}$$

3 Silenced Bias

In this section, we define what *silenced biases* are and how refusal steering helps reveal them. We find that most existing benchmarks often do not trigger refusals, even when bias is present. Moreover, we show that the refusal direction itself does not contain or cause social bias, and that refusal steering yields stable results.

Definition We define *silenced biases* as explicit biases extracted by QA prompts that the model initially refuses to comply with. Such biases are suppressed by LLMs' safety-alignment training, which obscures latent information. This builds upon the notion of *implicit bias* introduced by Bai et al. (2024), providing a complementary view on the underlying biases LLMs exhibit. Despite being suppressed at the output level, silenced biases can still influence the model's internal representations and decision-making processes. Critically,